

ABHISHEK SINGH

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Summary

ML Engineer with expertise in LLMs, RAG systems, agentic AI, and traditional ML, combined with hands-on **MLOps** and **AWS** deployment experience. Published **2 papers** in the top-tier **IEEE Transactions on Artificial Intelligence** journal and **1 IEEE ANTS** paper, and built scalable production AI pipelines.

Core Competencies: LLM Engineering, RAG Systems, Agentic AI, Model Optimization, MLOps, AWS Deployment

Technical Skills

Languages: Python, C/C++, SQL

ML/DL: PyTorch, TensorFlow, Transformers, RAG, LangChain, LangGraph

LLM/GenAI: SLMs, PEFT, LoRA, QLoRA, KV caching, Vector DBs (Chroma, FAISS)

MLOps: FastAPI, Docker, CI/CD (basic)

Tools: OpenCV, NLTK, Git

Databases: PostgreSQL, MongoDB

Cloud: AWS EC2 (deployment), S3 (storage), Lambda (async tasks), CloudWatch (monitoring)

Work Experience

Crosslynx

May 2025 – Present

Associate Data Scientist

On-Site

- Implemented a **RAG-based support chatbot** for smart meter installation queries, reducing repetitive support tickets and manual escalations by **48%**.
- Fine-tuned **Nomic embeddings**, improving retrieval accuracy and domain specific awareness (MAP@3 31%, Accuracy@5 28%).
- Integrated **Agentic memory mechanisms** to maintain multi-turn context, increasing historical recall success in field engineer conversations by **41%**.
- Deployed the entire pipeline on AWS (EC2, S3, Lambda) with CI/CD-enabled FastAPI microservices.
- Tech Stack:** Python, NLP, LangChain, LangFlow, Hugging Face, AWS

Crosslynx

Nov 2024 – May 2025

Data Science Intern

On-Site

- Developed an unsupervised **HDBSCAN-based electricity theft detection model** at the DTR level, identifying 5+ distinct theft clusters from consumption patterns.
- Outperformed baseline detection approaches, improving **AUC and MAP metrics by 38%** on DTR load datasets.
- Tech Stack:** Python, Statistics, A/B Testing, Unsupervised Algorithm, AWS

Quamin Agricare

August 2024 – October 2024

Data Science Intern

Remote

- Built a **soil health prediction model** using multivariate agronomy telemetry (moisture, NPK, pH, EC), improving **disease–nutrient diagnosis accuracy by 31%** through feature normalization and gradient-boosted regression pipelines.
- Applied **feature stability analysis, drift monitoring** to ensure robustness across soil types and crop cycles, reducing false nutrient imbalance predictions by **24%**.
- Tech Stack:** Python, Scikit-Learn, Gradient Boosting, Time-Series Forecasting, Feature Drift Analysis

Projects

Small Language Model | Pytorch, Generative AI, Transformer, Docker, Fast API [Git](#)

July 2024 – August 2024

- Developed a compact **Small Language Model (SLM)** with **16M parameters** using a **Transformer Decoder** architecture in **PyTorch**, capable of generating coherent English at a smaller scale compared to LLMs.
- Implemented **Key-Value (KV) Caching** to optimize auto-regressive inference by reducing redundant attention computations, significantly improving response latency.
- Containerized** the complete application using **Docker** to enable scalable and portable deployment across environments.

SecureKYC | Deep Learning, Computer Vision, Natural Language Processing [Git](#)

January 2024 – March 2024

- Integrated **Haarcascade**, **DeepFace** for face verification, and **EasyOCR** for ID text extraction, boosting KYC automation and accuracy by over 30%.
- Implemented **data hashing** and **MySQL** for secure storage, achieving a 50% increase in data privacy.

- Fine-tuned Llava using **PEFT** techniques on **Naver Clova IX CORD-v2 dataset** to prevent catastrophic loss and achieve **30% memory reduction**.
- Enabled structured JSON extraction from image receipts for efficient document analysis on **low-resource devices**.

- Engineered a custom email automation application using Meta's **LLaMA 3.1 model** and **Groq**, resulting in a **30% reduction** in manual client outreach.
- Integrated **LangChain WebLoader** and **ChromaDB** to **automate** job description analysis and semantic link extraction, enhancing email relevance by **25%**.

Publications

IEEE Top Q1 Journal

- **Attention-Guided Encoder-Decoder Design for Efficient RIS Phase Configuration in Time-Varying IoT Networks**

Accepted in *IEEE Transactions on Artificial Intelligence*, 2025 [\[Paper\]](#) [\[Code\]](#)

- Engineered an **attention-driven latent compression model** that learned high-salience control features, improving **phase-coherence retention by 42%** and reducing **information loss at 2/9–5/9 bit budgets by 37%**.
- Integrated **GDN/Inverse-GDN normalization** to stabilize distribution shift across noisy regimes, reducing **NMSE variance by 31%** under Gaussian and Rician interference.
- Benchmarked against PSC-DN, DL-CsiNet, and CsiNet, achieving **28–58% lower NMSE across compression ratios**, validating distribution-robust representation learning without model scaling or parameter inflation.

- **Bayesian Optimized Residual Convolutional Autoencoder for Efficient Feedback Compression in RIS-Assisted Time-Varying IoT Networks**

Accepted in *IEEE Transactions on Artificial Intelligence*, 2025 [\[Paper\]](#) [\[Code\]](#)

- Built a **Bayesian-optimized residual autoencoder** for RIS feedback compression, using **Optuna hyperparameter search** (depth, latent size, kernels) to cut **NMSE by 42% across SINR ranges**.
- Applied **LBFGS for final-epoch optimization**, stabilizing gradients under interference + fading and yielding **31% faster convergence** at low compression ratios.
- Operationalized tuning via **MLOps workflows** (trial tracking, pruning, reproducible search pipelines), outperforming prior methods under distribution shift.

International Conference

- **Federated Learning-Based Fault-Tolerant RIS Reconfiguration for Multi-User IoT Networks**

Accepted in *IEEE Advanced Networks and Telecommunications Systems (ANTS)*, 2025.

- Developed a **Federated Learning-driven RIS controller** for multi-user beam reconfiguration, enabling **distributed phase optimization** without centralized raw channel exchange.
- Implemented **FedAvg aggregation** across UE clusters to mitigate device/element failures and adversarial RIS malfunction, improving **sum-rate stability and SINR resilience** in heterogeneous environments.
- Benchmarked against brute-force and classical heuristics, achieving **near-optimal SINR and sum-rate at 62% lower compute cost**, validating scalable inference for high-frequency 6G deployments.

Education

Bachelor of Technology in Computer Science and Engineering

Indian Institute of Information Technology Bhagalpur

May 2021 – May 2025

CGPA: 8.35/10 (Absolute Grading)

- **Relevant Coursework:** Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Data Structures & Algorithms, Probability & Statistics, Linear Algebra, Optimization, Database Management Systems

Achievements

- Received a **Letter of Recommendation** for exceptional research contribution in the field of 6G communication from IIT BHU.
- Qualified twice at the regional level in the International Mathematics Olympiad (IMO).
- Serving as TPC member at **IEEE ANTS & ELEXCOM**, reviewing and evaluating AI research papers to ensure technical quality and relevance.
- Solved 700+ Leetcode problems, including 50 SQL questions, earning the SQL Badge.